

Time Series Analysis

Seminar 5: Conditional volatility: ARCH and GARCH



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Acronyms used in this material

- **ACF**: Autocorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **AR-GARCH**: AR model for the mean with GARCH errors
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **ARMA**: AutoRegressive Moving Average (model)
- **ARMA-GARCH**: ARMA model for the mean with GARCH errors
- **BET**: Bucharest Exchange Trading (index)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **DM**: Diebold–Mariano (test)
- **EGARCH**: Exponential GARCH
- **EODHD**: EOD Historical Data (market-data provider)
- **EWMA**: Exponentially Weighted Moving Average
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GJR**: Glosten–Jagannathan–Runkle (GARCH model)
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **IGARCH**: Integrated GARCH
- **LM**: Lagrange Multiplier
- **LR**: Likelihood Ratio (test)
- **MLE**: Maximum Likelihood Estimation
- **OLS**: Ordinary Least Squares
- **QLIKE**: Quasi-LIKElihood (loss function)
- **SE**: Standard Error
- **TSA**: Time Series Analysis

Today's Question and Seminar Structure

- The question of the seminar
  - **how do we detect, model, estimate and forecast a variance that changes every day?**
- Results to obtain (nothing is handed in)
  - the answers to the A exercises
  - a figure and a table for each B exercise
  - the corrected AI answer of C2
  - one sentence of interpretation for each exercise
- Notebook for today: open the seminar notebook in Google Colab
  - each task names its notebook section
- Exercises, their types and order: the exercise map (next slide)
- Seminar: prof. Daniel Traian Pele, danpele@ase.ro

Exercise map

Task	Question	Type	Model
A1, A2	persistence, long-run variance, half-life and one GARCH(1,1) step	Solved, Proposed	A1
A3, A4	multi-step forecasts, 10-day volatility and VaR 1%	Solved, Proposed	A3
A5, A6	an ARCH-LM test from regression output; news impact in GJR and EGARCH	Solved, Proposed	A5, A1
B1, B2	ARCH effects and AR(1)-GARCH(1,1)-t: BET; EUR/RON and Bitcoin	Solved, Proposed	B1
B3, B4	asymmetry and the leverage effect: S&P 500; BET, EUR/RON, Bitcoin	Solved, Proposed	B3
B5, B6	GARCH against EWMA out of sample, VaR 1%: S&P 500; BET, EUR/RON	Solved, Proposed	B5
C1, C2	is the leu calmer than it used to be? what is wrong in an AI answer?	Proposed	B1, A1

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model; the solution is discussed in class
- Part A: calculations on paper; Part B: real data in the notebook, each task ending with an interpretation question; Part C: open questions and the critique of an AI answer

Data used

Series	Source	Frequency	Period
S&P 500, BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Bitcoin	EODHD	daily, 7 days a week	2014–2026

- Daily log returns r_t in % (Prerequisites 1/21); weekends and repeated holiday closes are dropped (except Bitcoin)
- EUR/RON moves about 0.3% a day: the notebook estimates its models on $10\,r_t$ and converts the results back to %
- In the notebook: `returns('bet')`, `arch_lm(x)`, `fit(r, mean='AR', dist='t')` (package `arch`); no account or key is needed

Prerequisites for Today (1/21): returns, conditional mean and conditional variance

- **Daily log return** in %

$$r_t = 100 \ln \frac{P_t}{P_{t-1}}$$

  - P_t: the closing price (or the index level, the exchange rate) on day t
- **Mean and volatility model**: the return is a predictable part plus a scaled shock

$$r_t = \mu_t + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \text{ i.i.d., } E[z_t] = 0, \ \mathrm{Var}(z_t) = 1$$

  - $\mathcal{F}_{t-1}$: the information up to day $t-1$; $\mu_t = E[r_t \mid \mathcal{F}_{t-1}]$: the **conditional mean** (an ARMA model, Chapter 2)
  - $\sigma_t^2 = \mathrm{Var}(r_t \mid \mathcal{F}_{t-1})$: the **conditional variance**; σ_t: the conditional volatility
  - z_t: the **innovations**; i.i.d.: independent and identically distributed
- **Annualised volatility**: $\sqrt{252}\, s$ for daily returns
  - s: the sample standard deviation of r_t; 252: the number of trading days in a year

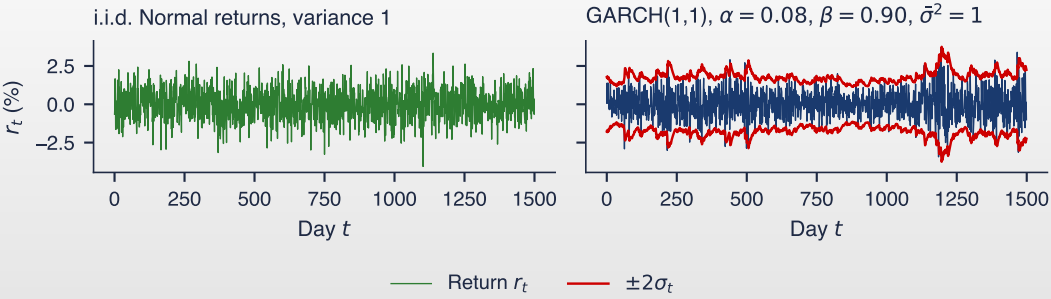
Prerequisites for Today (2/21): volatility clustering and kurtosis

- **Volatility clustering** Mandelbrot (1963): large changes follow large changes, small follow small
  - daily returns are almost uncorrelated, but $|r_t|$ and r_t^2 are positively autocorrelated
  - in the model: σ_t^2 moves slowly, so a turbulent day is followed by other turbulent days
- **Kurtosis**: the weight of the tails, measured by the fourth moment

$$K = \frac{E[(r_t - \mu)^4]}{\sigma^4}, \qquad \hat{K} = \frac{\frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^4}{\left(\frac{1}{T}\sum_{t=1}^{T}(r_t - \bar{r})^2\right)^2}$$

  - μ, σ^2: the unconditional mean and variance; $\bar{r}$: the sample mean; T: the number of days
  - $K = 3$ for the Normal distribution; $K > 3$: heavy tails, more extreme days than under the Normal distribution
  - a time-varying σ_t raises K even when z_t is Normal

Prerequisites for Today (3/21): a simulated GARCH series



- Left: i.i.d. Normal returns; right: GARCH(1,1) returns with the same long-run variance, 1
- Right: quiet and turbulent periods alternate; the band $\pm 2\sigma_t$ widens after large shocks
  - sample kurtosis: 3.07 (left) and 3.51 (right), although z_t is Normal in both

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Prerequisites for Today (4/21): ACF and the Ljung–Box test

- **Sample autocorrelation** at lag k

$$\hat{\rho}_k = \frac{\sum_{t=k+1}^{T}(x_t - \bar{x})(x_{t-k} - \bar{x})}{\sum_{t=1}^{T}(x_t - \bar{x})^2}$$

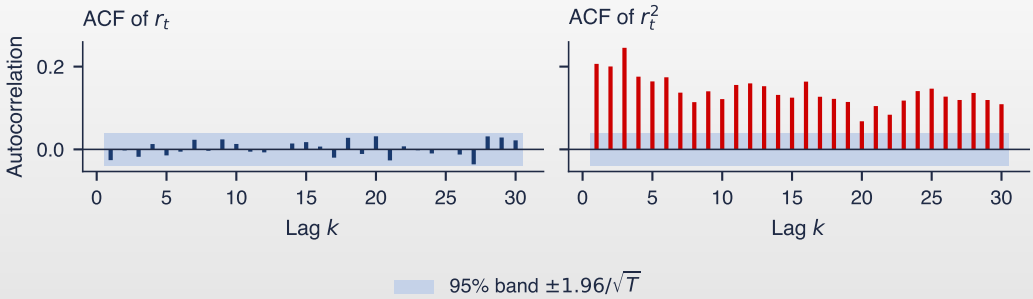
  - x_t: r_t, r_t^2 or the residuals; for white noise $\hat{\rho}_k \approx N(0, 1/T)$: 95% band $\pm 1.96/\sqrt{T}$

- **Ljung–Box** Ljung and Box (1978): the first m autocorrelations tested together

$$Q(m) = T(T+2)\sum_{k=1}^{m}\frac{\hat{\rho}_k^2}{T-k} \sim \chi^2(m)$$

  - H_0: $\rho_1 = \cdots = \rho_m = 0$; 5% value of $\chi^2(10)$: 18.31
  - on r_t: linear predictability of the returns; on r_t^2 or $\hat{\varepsilon}_t^2$: volatility clustering (ARCH effects)

Prerequisites for Today (5/21): ACF of returns and of squared returns



- The GARCH series of the previous slide, $T = 2\,500$ days; band $\pm 1.96/\sqrt{T} = \pm 0.039$
- r_t: inside the band, $Q(10) = 6.8$ below 18.31: no linear predictability
  - r_t^2: 0.21 at lag 1, still 0.12 at lag 10; $Q(10) = 746$: strong volatility clustering

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Prerequisites for Today (6/21): the ARCH-LM test

- **ARCH-LM** Engle (1982): an OLS regression of today's squared residual on the last q squares

$$\hat{\varepsilon}_t^2 = a_0 + a_1\hat{\varepsilon}_{t-1}^2 + \cdots + a_q\hat{\varepsilon}_{t-q}^2 + u_t$$

  - $\hat{\varepsilon}_t$: the residuals of the mean model (for example an AR(1)); a_j: the coefficients; u_t: the error
  - OLS: ordinary least squares; H_0: $a_1 = \cdots = a_q = 0$, no ARCH effects

- **Statistic**: the number of observations times the R^2 of the regression

$$\text{LM} = nR^2 \sim \chi^2(q), \qquad R^2 = 1 - \frac{RSS}{TSS}$$

  - n: the observations in the regression; RSS: the residual sum of squares; TSS: the total sum of squares of $\hat{\varepsilon}_t^2$
  - 5% values: $\chi^2(5)$ 11.07, $\chi^2(10)$ 18.31; LM above the value: reject H_0

Prerequisites for Today (7/21): the ARCH(1) model

- **ARCH(1)** Engle (1982): today's variance depends on yesterday's squared shock

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2, \qquad \omega > 0, \quad 0 \le \alpha < 1$$

  - ω: the minimum level of the variance; α: the reaction to yesterday's squared shock

- **Unconditional variance** and **kurtosis** (Normal z_t, $3\alpha^2 < 1$)

$$\mathrm{Var}(\varepsilon_t) = \frac{\omega}{1-\alpha}, \qquad K = \frac{3(1-\alpha^2)}{1-3\alpha^2} > 3$$

  - the larger α, the heavier the tails of ε_t, although z_t has the Normal distribution
  - a single lag is rarely enough: daily data need many lags of ε^2, hence GARCH

Prerequisites for Today (8/21): the GARCH(1,1) model

- **GARCH(1,1)** Bollerslev (1986): yesterday's variance enters as well

$$\sigma_t^2 = \omega + \alpha\,\varepsilon_{t-1}^2 + \beta\,\sigma_{t-1}^2, \qquad \omega > 0, \quad \alpha, \beta \geq 0$$

  - α: the reaction to the newest shock; β: the memory, the weight of yesterday's variance
- **Persistence** $\alpha + \beta$ and **long-run variance** (if $\alpha + \beta < 1$)

$$\bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}, \qquad \text{annual volatility} = \sqrt{252\,\bar{\sigma}^2}$$

  - $\alpha + \beta < 1$: covariance stationary, σ_t^2 reverts to $\bar{\sigma}^2$; close to 1: shocks last long
- **Half-life**: the number of days in which half of a variance shock disappears

$$h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$$

  - from $(\alpha + \beta)^h = 0.5$; $\alpha + \beta = 0.97$: about 23 days

Prerequisites for Today (9/21): variance forecasts

- Forecasts made at the end of day t, with $E_t[\cdot] = E[\cdot \mid \mathcal{F}_t]$

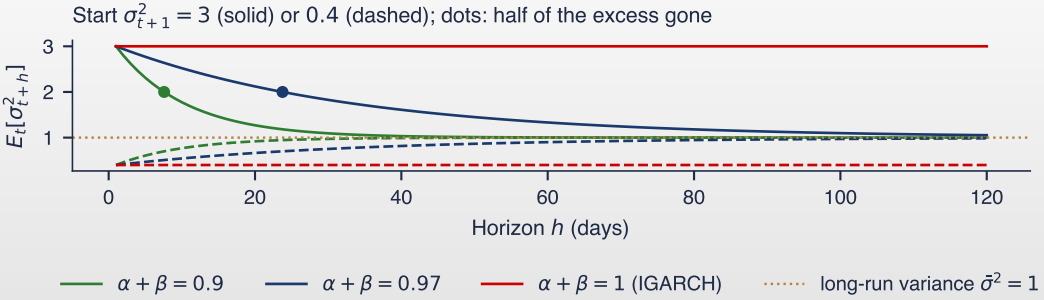
$$\sigma^2_{t+1} = \omega + \alpha(r_t - \mu)^2 + \beta\sigma^2_t, \qquad E_t[\sigma^2_{t+h}] = \bar{\sigma}^2 + (\alpha+\beta)^{h-1}(\sigma^2_{t+1} - \bar{\sigma}^2)$$

  - μ: the constant mean; tomorrow's variance is known today; the gap to $\bar{\sigma}^2$ shrinks by the factor $\alpha+\beta$ per day
- **Variance over H days**: the sum of the daily forecasts

$$\sum_{h=1}^{H} E_t[\sigma^2_{t+h}] = H\bar{\sigma}^2 + (\sigma^2_{t+1} - \bar{\sigma}^2)\,\frac{1-(\alpha+\beta)^H}{1-\alpha-\beta}$$

  - the square-root-of-time rule $\sqrt{H}\,\sigma_{t+1}$ assumes the variance stays at σ^2_{t+1}; $\sqrt{H\bar{\sigma}^2}$ assumes it is already at $\bar{\sigma}^2$

Prerequisites for Today (10/21): mean reversion of the forecasts



- $E_t[\sigma^2_{t+h}]$ for $\bar{\sigma}^2 = 1$, starting from 3 (after a shock) or from 0.4 (a calm period)
- Half-life: 6.6 days for $\alpha + \beta = 0.90$, 22.8 days for 0.97 (dots)
  - IGARCH, $\alpha + \beta = 1$: the forecast never returns, every horizon repeats σ^2_{t+1}

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Prerequisites for Today (11/21): EWMA, IGARCH and AR-GARCH

- **EWMA** J.P. Morgan/Reuters (1996): an exponentially weighted moving average of r_t^2 (Chapter 0)

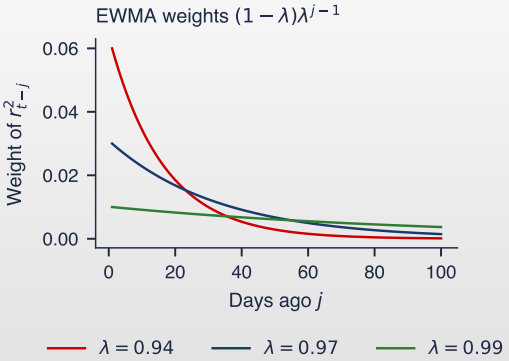
$$\sigma_t^2 = \lambda\sigma_{t-1}^2 + (1-\lambda)\, r_{t-1}^2 = (1-\lambda)\sum_{j=1}^{\infty}\lambda^{j-1} r_{t-j}^2$$

  - λ: the weight of yesterday's variance, 0.94 for daily data in RiskMetrics; no parameter to estimate
- **IGARCH** (integrated GARCH): $\alpha + \beta = 1$
  - EWMA is an IGARCH with $\omega = 0$: no long-run variance and no half-life
- **AR(1)-GARCH(1,1)**: an AR(1) for the mean, GARCH errors, estimated together

$$r_t = c + \phi\, r_{t-1} + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

  - c: the constant; ϕ: the AR(1) coefficient; **-t** in GARCH-t: Student-t innovations

Prerequisites for Today (12/21): the weights of EWMA



- Weight of the squared return of j days ago
  - $(1-\lambda)\lambda^{j-1}$; the weights sum to 1
- Share of the last 20 days
  - $\lambda = 0.94$: 71%
  - $\lambda = 0.97$: 46%
  - $\lambda = 0.99$: 18%
- A smaller λ
  - reacts faster, but is noisier

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Prerequisites for Today (13/21): value at risk

- **VaR 1%** (value at risk) for tomorrow: the loss exceeded with probability 1%

$$P(r_{t+1} < -\mathrm{VaR}_{t+1} \mid \mathcal{F}_t) = 0.01, \qquad \mathrm{VaR}_{t+1} = -\big(\mu + \sigma_{t+1}\, q_{0.01}(z)\big)$$

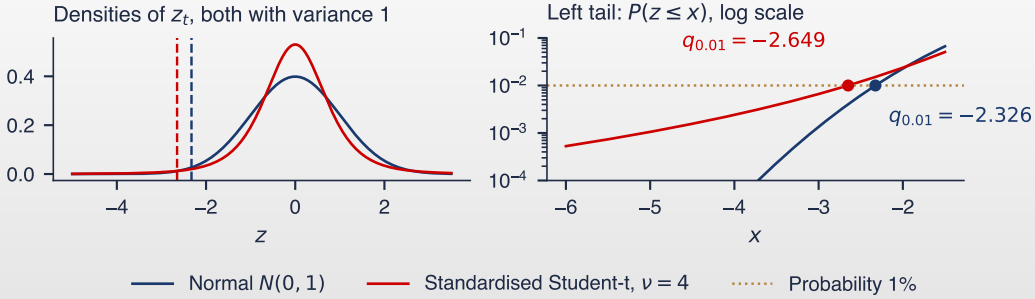
  - $q_{0.01}(z)$: the 1% quantile of the innovation z, negative; VaR is reported as a positive loss in %
  - over H days: σ_{t+1} is replaced by the square root of the H-day variance

- **Quantiles**: Normal, and Student-t rescaled to variance 1 Bollerslev (1987)

$$q_{0.01} = -2.326 \text{ (Normal)}, \qquad q_{0.01} = t_\nu^{-1}(0.01)\sqrt{\frac{\nu-2}{\nu}} \text{ (Student-t)}$$

  - $\nu > 2$: the degrees of freedom, a smaller ν means heavier tails; t_ν^{-1}: the quantile function of $t(\nu)$
  - $\mathrm{Var}(t_\nu) = \nu/(\nu-2)$, hence the factor $\sqrt{(\nu-2)/\nu}$; example: $t_5^{-1}(0.01) = -3.365$

Prerequisites for Today (14/21): the 1% quantile with heavy tails



- Both innovations have variance 1; the Student-t with $\nu = 4$ has a higher peak and heavier tails
- 1% quantile: -2.326 (Normal), -2.649 (Student-t, $\nu = 4$)
  - at the same σ_{t+1}, the Student-t VaR 1% is 14% larger

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Prerequisites for Today (15/21): maximum likelihood and model comparison

- **MLE** (maximum likelihood estimation), Normal innovations

$$\ell(\boldsymbol{\vartheta}) = -\frac{1}{2}\sum_{t=1}^{T}\left[\ln 2\pi + \ln \sigma_t^2 + \frac{(r_t - \mu_t)^2}{\sigma_t^2}\right]$$

  - $\boldsymbol{\vartheta}$: all parameters (c, ϕ, ω, α, β, and ν for Student-t); $\hat{\boldsymbol{\vartheta}}$ maximises ℓ
  - robust SE (standard errors) Bollerslev and Wooldridge (1992): valid even when z_t is not Normal
- **LR test** (likelihood ratio) of a smaller model against a larger one

$$\text{LR} = 2(\ell_1 - \ell_0) \sim \chi^2(r), \qquad \text{BIC} = -2\ell + k \ln T$$

  - ℓ_1, ℓ_0: the maximised log-likelihoods of the larger and of the smaller model; r: the number of restrictions; $\chi^2_{0.95}(1) = 3.84$
  - BIC Schwarz (1978): k parameters, T observations; the smaller BIC wins; it penalises extra parameters more than LR

Prerequisites for Today (16/21): asymmetric GARCH models

- **GJR-GARCH** Glosten, Jagannathan and Runkle (1993): a fall adds the extra reaction γ

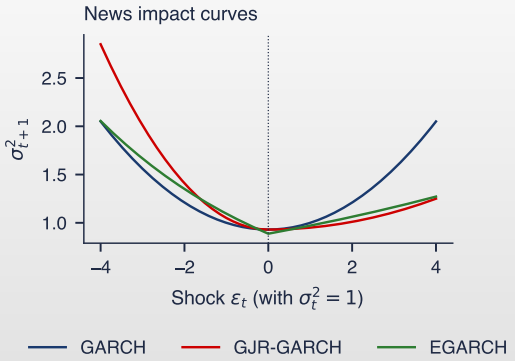
$$\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\,\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \qquad I_{t-1} = \mathbf{1}\{\varepsilon_{t-1} < 0\}$$

  - $\mathbf{1}\{\cdot\}$: 1 if the condition holds, 0 otherwise; **leverage effect** Christie (1982): $\gamma > 0$, falls raise volatility more
  - persistence $\alpha + \beta + \gamma/2$ (a fall happens half of the time)
- **EGARCH** Nelson (1991): a model for $\ln\sigma_t^2$, always positive

$$\ln\sigma_t^2 = \omega + \alpha\big(|z_{t-1}| - E|z_{t-1}|\big) + \gamma z_{t-1} + \beta\ln\sigma_{t-1}^2$$

  - α: the size effect; γ: the sign effect, leverage if $\gamma < 0$; $E|z| = \sqrt{2/\pi} = 0.7979$ for Normal z
- **News impact curve**: σ_{t+1}^2 as a function of today's shock ε_t, with σ_t^2 fixed

Prerequisites for Today (17/21): news impact curves



- $\sigma_t^2 = 1$; same persistence
  - GARCH: $\alpha = 0.07$, $\beta = 0.90$
  - GJR: $\alpha = 0.02$, $\gamma = 0.10$, $\beta = 0.90$
  - EGARCH: $\alpha = 0.15$, $\gamma = -0.06$, $\beta = 0.97$
- GARCH is symmetric: 1.21 after ± 2
- GJR: 1.41 after -2, 1.01 after $+2$
  - EGARCH: 1.35 against 1.06

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Prerequisites for Today (18/21): the sign-bias test

- **Standardised residuals**: the estimated innovations

$$\hat{z}_t = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t}$$

  - for a correct model, $\hat{z}_t$ and $\hat{z}_t^2$ have no autocorrelation (Ljung–Box)

- **Sign-bias test** Engle and Ng (1993): can the sign of yesterday's shock predict $\hat{z}_t^2$?

$$\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{z}_{t-1} + b_3 S_{t-1}^+ \hat{z}_{t-1} + e_t$$

  - $S_{t-1}^- = \mathbf{1}\{\hat{z}_{t-1} < 0\}$, $S_{t-1}^+ = 1 - S_{t-1}^-$; b_1: the sign bias, b_2, b_3: the size bias of falls and rises
  - joint test H_0: $b_1 = b_2 = b_3 = 0$, statistic $nR^2 \sim \chi^2(3)$; a rejection: the model misses an asymmetry

Prerequisites for Today (19/21): comparing variance forecasts

- **Out of sample**: the forecast h_t for day t uses only data up to $t-1$
  - the model is re-estimated from time to time (every 250 days) on all data up to that day; r_t^2: a noisy proxy of the true variance
- **QLIKE** loss Patton (2011)

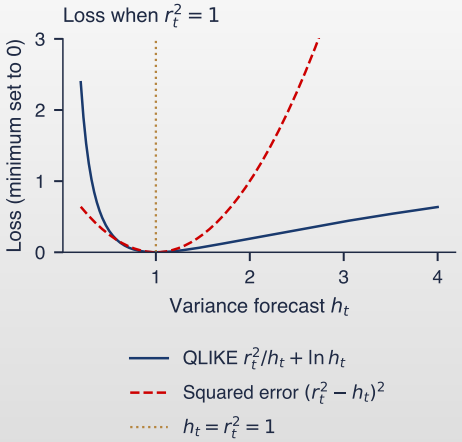
$$L_t = \frac{r_t^2}{h_t} + \ln h_t$$

  - minimal at $h_t = r_t^2$; robust to the noise of the proxy; a smaller mean QLIKE is better
- **DM test** Diebold and Mariano (1995): models A and B, loss differential $d_t = L_t^A - L_t^B$

$$\mathrm{DM} = \frac{\bar{d}}{\mathrm{SE}_{\mathrm{HAC}}(\bar{d})} \approx N(0,1)$$

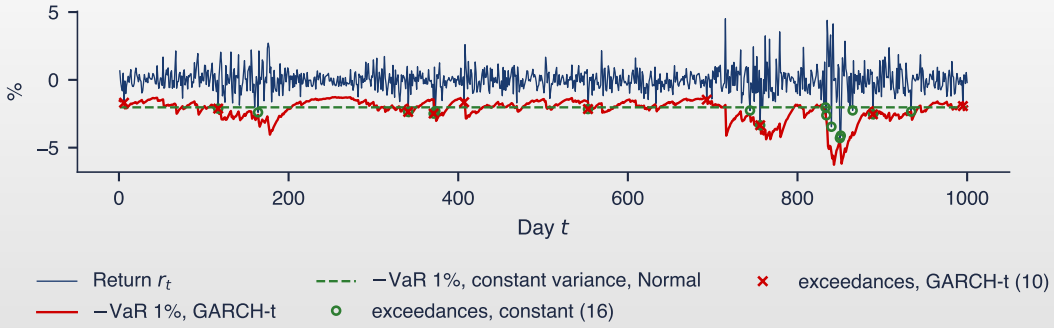
  - $\bar{d}$: the mean of d_t; HAC: a standard error robust to heteroskedasticity and autocorrelation of d_t
  - DM < -1.96: A significantly better at 5%; > 1.96: B better

Prerequisites for Today (20/21): the QLIKE loss



- $r_t^2 = 1$; both losses are 0 at $h_t = 1$
- QLIKE: under-prediction costs more
  - $h_t = 0.5$: 0.31; $h_t = 2$: 0.19
- Squared error: the reverse
  - $h_t = 0.5$: 0.25; $h_t = 2$: 1.00
  - dominated by a few days with huge r_t^2
- A risk manager fears under-prediction
  - so QLIKE suits risk models

Prerequisites for Today (21/21): VaR exceedances



- **Exceedance**: a day with $r_t < -\text{VaR}_t$; a correct VaR 1% is exceeded on about 1% of the n days
  - simulated GARCH-t returns, $n = 1\,000$, 10 exceedances expected: GARCH-t VaR 10, constant Normal VaR 16, grouped in the turbulent months

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A1: GARCH(1,1) algebra [Solved]

Task

- Daily returns in %: $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, $\mu = 0$, 252 trading days per year.
- 1. Compute the persistence and the long-run variance.
- 2. Compute the annualised long-run volatility.
- 3. Compute the half-life of a variance shock.
- 4. Today $\sigma_t^2 = 1.5$ and $r_t = -3\%$; compute σ_{t+1}^2 and σ_{t+1}.
- Report: four numbers and one sentence.

Solution

- 1. $\alpha + \beta = 0.98$; $\bar{\sigma}^2 = 0.02/(1 - 0.98) = 1.000$
- 2. $\sqrt{252 \times 1.000} = 15.87\%$ per year
- 3. $\ln 0.5 / \ln 0.98 = -0.69315/(-0.02020) = 34.3$ days
- 4. $0.02 + 0.08 \times 9 + 0.90 \times 1.5 = 2.090$; $\sigma_{t+1} = 1.446\%$
- A fall of 3% raises the variance by 39%; half of the excess is gone after 34.3 trading days, about 7 weeks.

A2: a more reactive market [Proposed]

Solution discussed in class

Task

- $\omega = 0.05$, $\alpha = 0.12$, $\beta = 0.85$, $\mu = 0$; today $\sigma_t^2 = 2.0$ and $r_t = +1.5\%$. Model: A1.
- 1. Compute the persistence, the long-run variance and the annualised long-run volatility.
- 2. Compute the half-life.
- 3. Compute σ_{t+1}^2.
- 4. Write EWMA with $\lambda = 0.94$ as a GARCH(1,1) and say which of the quantities in 1–2 exist for it.
- Report: four numbers and two sentences.

Back Exercise list

A3: forecasts and VaR 1% [Solved]

Task

- The model of A1, with $\sigma^2_{t+1} = 2.090$ and Normal innovations.
- 1. Compute $E_t[\sigma^2_{t+5}]$ and $E_t[\sigma^2_{t+10}]$.
- 2. Compute the 10-day variance and volatility.
- 3. Compare with the square-root-of-time rule $\sqrt{10}\,\sigma_{t+1}$ and with $\sqrt{10\,\bar{\sigma}^2}$.
- 4. Compute the one-day and the 10-day VaR 1%.
- Report: six numbers and one sentence.

Solution

- 1. $1 + 0.98^4(2.090 - 1) = 1 + 0.9224 \times 1.09 = 2.005$; $1 + 0.98^9 \times 1.09 = 1.909$
- 2. $10 \times 1 + 1.09 \times (1 - 0.98^{10})/0.02 = 10 + 1.09 \times 9.146 = 19.97$; volatility 4.47%
- 3. $\sqrt{10 \times 2.090} = 4.57\%$ (too high), $\sqrt{10} = 3.16\%$ (too low)
- 4. $2.326 \times 1.446 = 3.36\%$; $2.326 \times 4.47 = 10.40\%$
- The high variance decays slowly: the 10-day risk lies between the two simple rules, closer to the first.

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A4: forecasts with Student-t innovations [Proposed]

Solution discussed in class

Task

- The model of A2, with $\sigma^2_{t+1} = 2.020$ and standardised Student-t innovations with $\nu = 5$ ($t_5^{-1}(0.01) = -3.365$). Model: A3.
- 1. Compute $E_t[\sigma^2_{t+10}]$ and the 10-day variance.
- 2. Compute the quantile $q_{0.01}$ of the standardised t.
- 3. Compute the one-day and the 10-day VaR 1% with t and with Normal innovations.
- Report: six numbers and one sentence.

A5: an ARCH-LM test from regression output [Solved]

Task

- An AR(1) is fitted to the daily returns of a stock index. The regression of $\hat{\varepsilon}_t^2$ on a constant and five lags uses $n = 1000$ observations and gives $R^2 = 0.062$; the Ljung–Box statistic of $\hat{\varepsilon}_t^2$ is $Q(10) = 95.3$.
- 1. Compute the ARCH-LM statistic and compare it with the 5% value of $\chi^2(5)$.
- 2. Compare $Q(10)$ with the 5% value of $\chi^2(10)$.
- 3. Say which model you would fit next.
- 4. For an ARCH(1) with $\omega = \alpha = 0.5$ and Normal z_t, compute the unconditional variance and the kurtosis.
- Report: two decisions, two numbers and one sentence.

Solution

- 1. LM $= 1000 \times 0.062 = 62.0 > 11.07$ ($p < 0.001$): reject H_0, ARCH effects
- 2. $95.3 > 18.31$ ($p < 0.001$): the squares are autocorrelated; same conclusion
- 3. AR(1)-GARCH(1,1), estimated jointly; then the same tests on $\hat{z}_t$
- 4. $\bar{\sigma}^2 = 0.5/(1 - 0.5) = 1.0$; $K = 3 \times 0.75/0.25 = 9.0$: Normal innovations, yet heavy-tailed returns

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A6: good and bad news [Proposed]

Solution discussed in class

Task

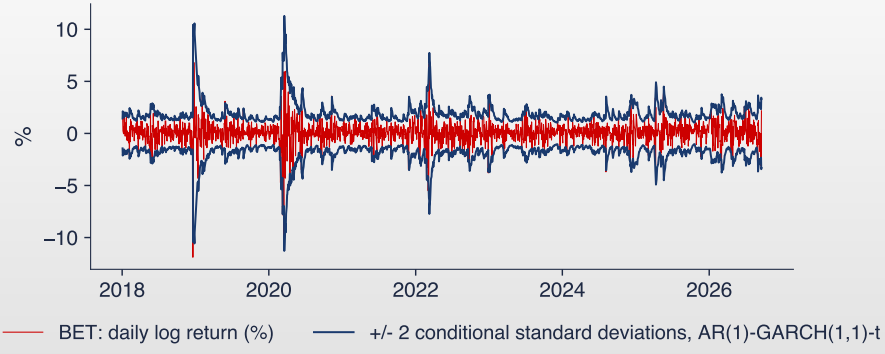
- GJR-GARCH: $\omega = 0.02$, $\alpha = 0.03$, $\gamma = 0.12$, $\beta = 0.90$; EGARCH: $\omega = 0$, $\alpha = 0.12$, $\gamma = -0.08$, $\beta = 0.98$; in both $\sigma_t^2 = 1$, $\mu = 0$. Model: A1.
- 1. Compute the GJR σ_{t+1}^2 after $\varepsilon_t = -2$ and after $\varepsilon_t = +2$, and their ratio.
- 2. Compute the GJR persistence and half-life.
- 3. Compute the EGARCH σ_{t+1}^2 for the same two shocks (use $E|z| = 0.7979$).
- Report: six numbers and one sentence.

B1: ARCH effects and AR(1)-GARCH(1,1)-t for the BET [Solved]

- **Question**: does the BET have ARCH effects, how persistent is its volatility, and where is it today relative to its long-run level?
- **Inputs**: BET closes since 2000, daily log returns in %
- **Tasks**
  1. Fit an AR(1) for the mean by OLS, then run ARCH-LM(5) and Ljung–Box $Q(10)$ on its squared residuals.
  2. Estimate AR(1)-GARCH(1,1)-t and report the parameters with robust standard errors.
  3. Compute the persistence, the half-life and the annualised long-run volatility, and compare them with the sample volatility.
  4. Check $Q(10)$ of $\hat{z}_t$ and of $\hat{z}_t^2$, and draw the returns since 2018 with $\pm 2\hat{\sigma}_t$ bands.
  5. Interpretation: is the BET volatility on the last day above or below its long-run level, and what does the model expect for the next months?
- **Report**: two test statistics, a table of six parameters, five numbers, the chart and two sentences
- Notebook: section B1

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B1: solution (1/2) [Solved]



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B1: solution (2/2) [Solved]

	c	ϕ	ω	α	β	ν
estimate	0.077	0.091	0.0441	0.189	0.801	5.09
robust SE	0.010	0.013	0.0085	0.023	0.023	0.32

- AR(1) residuals: ARCH-LM $= nR^2 = 1017$ ($R^2 = 0.153$), $Q(10)$ of squares $= 2496$: strong ARCH effects
- $T = 6\,669$; $\alpha + \beta = 0.990$, half-life $\ln 0.5/(-0.01028) = 67.4$ days; long-run volatility 32.8% against 22.1% in the sample; $\hat{z}_t$: $Q(10) = 28.6$ ($p < 0.001$), $\hat{z}_t^2$: 7.2 ($p = 0.705$)
- Peaks: 114% (2008), 89% (2020); last day: 26.2%; 5.1% of the days fall outside $\pm 2\hat{\sigma}_t$
- Interpretation: below (26.2% against 32.8%); the model expects the volatility to rise slowly towards the long-run level, but that level is imprecise ($1 - \alpha - \beta = 0.0102$)

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B2: EUR/RON and Bitcoin [Proposed]

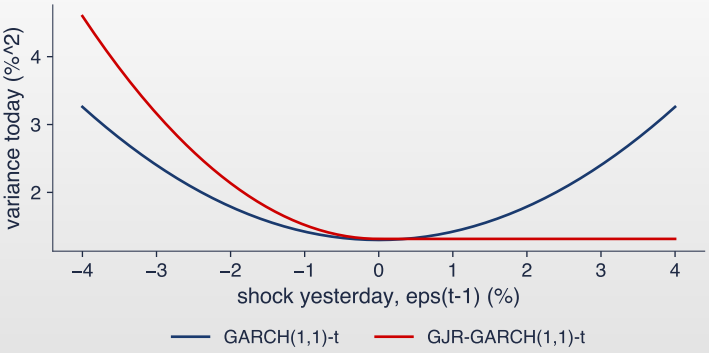
- **Question**: do the EUR/RON rate and Bitcoin behave like the BET? Model: B1.
- **Inputs**: EUR/RON (BNR reference rate) since July 2005, Bitcoin since September 2014; daily log returns in %
- **Tasks**
  1. Repeat steps 1–4 of B1 for both series (EUR/RON: estimate on $10\, r_t$).
  2. Report $\hat{\alpha} + \hat{\beta}$ and say whether the half-life and the long-run volatility exist.
  3. Interpretation: what does $\hat{\alpha} + \hat{\beta} = 1$ mean for a forecast of the volatility one year ahead?
- **Report**: one table and two sentences
- Notebook: section B2

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B3: the leverage effect in the S&P 500 [Solved]

- **Question**: do falls raise the volatility of the S&P 500 more than rises of the same size?
- **Inputs**: S&P 500 closes since 2000, daily log returns in %
- **Tasks**
  1. Estimate GARCH(1,1)-t and GJR-GARCH(1,1)-t; report γ with its robust t statistic.
  2. Compute the LR statistic of GJR against GARCH, its p-value and the two BIC values.
  3. Run the sign-bias test on the standardised residuals of GARCH-t.
  4. Draw both news impact curves and compute the GJR variance after shocks of -2% and $+2\%$.
  5. Interpretation: how does the leverage effect change the VaR on the day after a fall?
- **Report**: four statistics with p-values, two variances, the chart and two sentences
- Notebook: section B3

B3: solution (1/2) [Solved]



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B3: solution (2/2) [Solved]

- GJR: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$; LR $= 235.7$ ($p < 0.001$); BIC 18169.0 $\rightarrow$ 17942.2
- Sign bias on the GARCH-t residuals: $\chi^2(3) = 40.7$ ($p < 0.001$), sign $t = 3.56$
- GJR variance after -2%: 2.14; after $+2\%$: 1.31; ratio 1.62
- Interpretation: only falls raise the variance ($\hat{\alpha} = 0$); after a fall the VaR is about $\sqrt{1.62} = 1.27$ times the VaR after an equal rise; a symmetric GARCH understates the risk after bad days

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B4: asymmetry in the BET, EUR/RON and Bitcoin [Proposed]

- **Question**: is there a leverage effect in the BET, in EUR/RON and in Bitcoin? Model: B3.
- **Inputs**: BET since 2000, EUR/RON since 2005, Bitcoin since 2014; daily log returns in %
- **Tasks**
  1. Estimate GJR-GARCH(1,1)-t and EGARCH(1,1)-t; report both values of γ with robust t statistics.
  2. Compute the LR test of GJR against GARCH and the change in BIC.
  3. Interpretation: for EUR/RON a positive return means a weaker leu; what is "bad news" for this series?
- **Report**: one table and two sentences
- Notebook: section B4

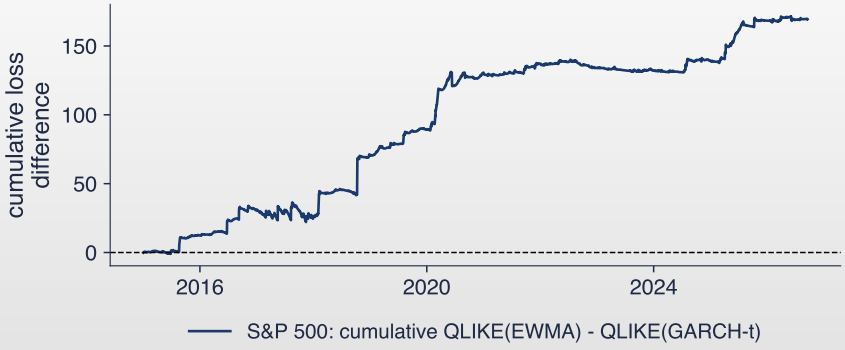
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B5: GARCH against EWMA for the S&P 500 [Solved]

- **Question**: does GARCH(1,1)-t forecast tomorrow's variance of the S&P 500 better than EWMA, and is its VaR 1% exceeded on 1% of the days?
- **Inputs**: S&P 500 since 2000; forecasts for every day since 2 January 2015
- **Tasks**
  1. Re-estimate GARCH(1,1)-t every 250 days on all data up to that day and compute the one-day forecasts h_t; compute EWMA with $\lambda = 0.94$.
  2. Compute the mean QLIKE of both models and the Diebold–Mariano t statistic.
  3. Count the exceedances of the GARCH-t VaR 1% and of the EWMA-Normal VaR 1%, and compare them with the expected number.
  4. Draw the cumulative difference of the losses.
  5. Interpretation: which model would you give to a risk manager, and what would you still improve?
- **Report**: four numbers, two counts, the chart and two sentences
- Notebook: section B5

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B5: solution (1/2) [Solved]



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B5: solution (2/2) [Solved]

- 2 944 days; mean QLIKE: GARCH-t 0.761, EWMA 0.819; DM: mean difference -0.058, $t = -3.69$ ($p < 0.001$): GARCH-t is better
- VaR 1% exceedances: GARCH-t 49 (1.66%), EWMA-Normal 65 (2.21%); expected about 29
- Interpretation: GARCH-t, but both VaR models are exceeded too often; add the leverage effect (B3) and test the number of exceedances formally (Kupiec (1995))

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B6: GARCH against EWMA for the BET and EUR/RON [Proposed]

- **Question**: does the result of B5 hold for the BET and for EUR/RON? Model: B5.
- **Inputs**: BET since 2000, EUR/RON since 2005; forecasts for every day since January 2015
- **Tasks**
  1. Compute the out-of-sample GARCH(1,1)-t and EWMA forecasts as in B5.
  2. Compute the mean QLIKE of both models and the DM test.
  3. Find the day with the largest loss difference, and recompute the mean QLIKE without it.
  4. Count the VaR 1% exceedances of both models.
  5. Interpretation: why can the DM test be insignificant when one mean QLIKE is half the other?
- **Report**: one table and two sentences
- Notebook: section B6

C1: is the leu calmer than it used to be? [Proposed]

- **Question**: has the volatility of EUR/RON changed between 2005–2012, 2013–2019 and 2020–2026, and can one GARCH model describe all three periods?
- **Inputs**: EUR/RON daily log returns in % (BNR reference rate); model: B1
- **Tasks**
  1. For each period compute the annualised sample volatility and the share of days with $|r_t| > 0.5\%$.
  2. Estimate GARCH(1,1)-t on each period and report $\alpha + \beta$, α and ν.
  3. Find the largest daily move of each period and its date.
  4. Interpretation: is the change in volatility a property of the market, or of the exchange-rate policy?
- **Report**: a table and a plan for a project
- Notebook: section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to interpret a GARCH(1,1)-t fit of the BET (daily, since 2000): $\hat{\omega} = 0.0448$, $\hat{\alpha} = 0.192$, $\hat{\beta} = 0.799$, $\hat{\nu} = 5.01$. The answer:
- `(a) The half-life of a volatility shock is ln 0.5 / ln beta = 3.1 days.`
- `(b) The long-run variance is omega/(1 - alpha) = 0.055, an annual volatility of 3.7%.`
- `(c) An AR(1) by OLS gives phi = 0.109 with SE 0.012, t = 8.9: BET returns are strongly predictable.`
- `(d) Ljung-Box Q(10) of the squared standardised residuals = 8.3, p = 0.60: the GARCH-t model is correct.`
- `(e) With Student-t innovations the 1% quantile of z is closer to 0 than the Normal -2.326, so the VaR 1% is smaller.`
- `(f) alpha + beta = 0.991 < 1, so the model is covariance stationary and the volatility reverts to its long-run level.`
- Tasks
  - 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - 2. Report: a list of six verdicts with one line of justification each.

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Key takeaways

- Test for ARCH effects on the residuals of the mean model: ARCH-LM $= nR^2$ and Ljung–Box on the squares
- $\alpha + \beta$ decides the long run: half-life, long-run variance, the shape of the forecasts; $\alpha + \beta = 1$: IGARCH
- Multi-day risk is the sum of the daily variance forecasts, not $\sqrt{H}$ times today's volatility
- Heavy-tailed innovations change the VaR quantile; asymmetry changes the VaR after bad days
- Compare forecasts out of sample (QLIKE, DM) and look for single days that drive the result

After the seminar

- Details: Chapter 5: stylised facts, ARCH and GARCH, estimation, ARMA-GARCH, asymmetry, diagnostics, forecasts, VaR 1%
- C1 can grow into a team project: volatility regimes of the leu and of other currencies of the region
- Reading: Huang and Petukhina (2022), Ch. 6; Tsay (2010), Ch. 3

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